1.15 Graph Sketching: Cubic Graphs

Chapter 4 · Original slide 4

Polynomial Graphs

4

In Chapter 2 we briefly saw that a **polynomial** expression is of the form:

$$a + bx + cx^2 + dx^3 + ex^3 + \dots$$

where $a, b, c, d, e, \dots$ are constants (which could be 0).

The **order** of a polynomial is its highest power.

Order	Name
0	Constant (e.g. “4”)
1	
2	
3	
4	
5	

These are covered in Chapter 5.

Chapter 2 explored the graphs for these.

We will cover these now.

While these are technically beyond the A Level syllabus, we will look at how to sketch polynomials in general.

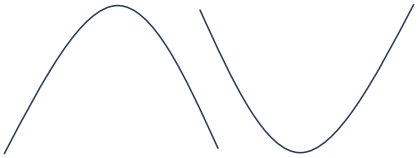
Chapter 4 · Original slide 5

Polynomial Graphs

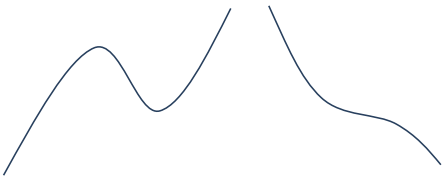
5

Order:

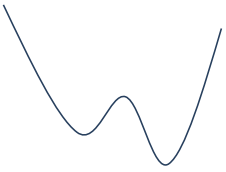
2



3



4



What property connects the order of the polynomial and the shape?

In Chapter 2 how did we tell what way up a quadratic is, and why does this work?

1.15 Graph Sketching: Cubic Graphs

Chapter 4 · Original slide 6

Polynomial Graphs

6

e.g. If $y = 2x^2 + 3$, try a large positive value like $x = 1000$. We can see we'd get a large positive y value. Thus as $x \to \infty$, $y \to \infty$

Equation	If $a > 0$	Resulting Shape	If $a < 0$	Resulting Shape
$y = ax^2 + bx + c$	As $x \to \infty, y \to \infty$ As $x \to -\infty, y \to \infty$		As $x \to \infty, y \to -\infty$ As $x \to -\infty, y \to -\infty$	
$y = ax^3 + bx^2 + cx + d$				
$y = ax^4 + bx^3 + cx^2 + dx + e$				
$y = ax^5 + bx^4 + \ldots$				

If $a > 0$, what therefore can we say about the shape if:

- **The order is odd**: It goes
- **The order is even**: The ta

Chapter 4 · Original slide 7

Cubics

7

Sketch the curve with equation
$y = (x - 2)(1 - x)(1 + x)$

Features you must consider:

Shape?

Roots?

y-intercept?

Sketch the curve with equation
$y = x^2(x - 1)$

Shape?

Roots?

y-intercept?

1.15 Graph Sketching: Cubic Graphs

Chapter 4 · Original slide 8

Cubics

8

Sketch the curve with equation
$y = (2 - x)(x + 1)^2$

Shape?

Roots?

y-intercept?

Sketch the curve with equation
$y = (x - 4)^3$

Shape?

Roots?

y-intercept?

Chapter 4 · Original slide 9

Cubics with Limited Roots

9

Sketch the curve with equation
$y = (x + 1)\left(x^2 + x + 1\right)$

Shape?

Roots?

y-intercept?

1.15 Graph Sketching: Cubic Graphs

Chapter 4 · Original slide 10

Finding the equation yourself

10

Edexcel C1 May 2013(R)
Q9

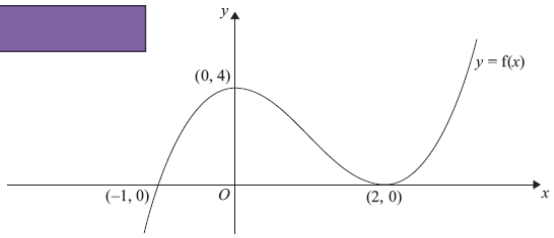


Figure 1 shows a sketch of the curve C with equation $y = f(x)$.
The curve C passes through the point (–1, 0) and touches the x-axis at the point (2, 0).
The curve C has a maximum at the point (0, 4).
The equation of the curve C can be written in the form.

$$y = x^3 + ax^2 + bx + c$$

where a, b and c are integers.
(a) Calculate the values of a, b, c.

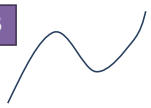
Chapter 4 · Original slide 11

Test Your Understanding

11

1 Sketch the curve with equation
$y = x(x-3)^2$

2 Sketch the curve with equation
$y = -(x+2)^3$

3 

A curve has this shape, touches the x axis at 3 and crosses the x axis at -2. Give a suitable equation for this graph.

☠ Sketch the curve with equation
$y = 2x^2(x-1)(x+1)^3$

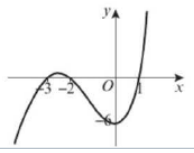
1.15 Graph Sketching: Cubic Graphs

Chapter 4 · Original slide 12

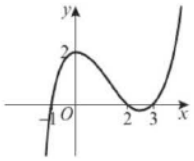
Exercise 4A –Y1 Pure pages 63

12

5 The graph of $y = x^3 + bx^2 + cx + d$ is shown opposite, where b, c, d are real constants.
a Find the values of b, c, d. (3 marks)
b Write down the coordinates of the point where the curve crosses the y-axis. (1 mark)



6 The graph of $y = ax^3 + bx^2 + cx + d$ is shown opposite, where a, b, c, d are real constants. Find the values of a, b, c, d. (4 marks)



Chapter 4 · Original slide 13

Exercise 4A –Y1 Pure pages 63

13

7 Given that $f(x) = (x - 10)(x^2 - 2x) + 12x$

a Express $f(x)$ in the form $x(ax^2 + bx + c)$ where a, b, c are real constants.
b Hence factorise $f(x)$ completely.
c Sketch the graph of $y = f(x)$ showing clearly the points where the graph intersects the axes.

1.15 Graph Sketching: Cubic Graphs

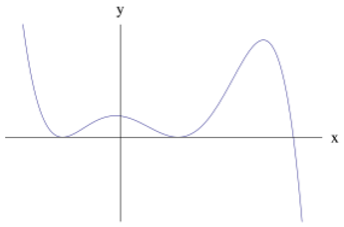
Chapter 4 · Original slide 14

Exercise 4A Extension

14

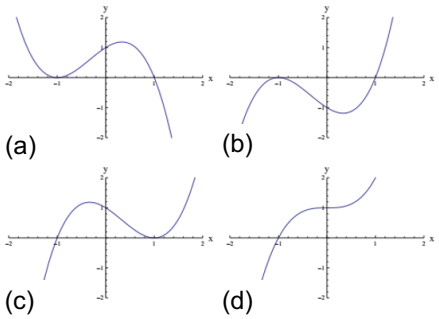
Extension

1 [MAT 2012 1E] Which one of the following equations could possibly have the graph given below?



A) $y = (3-x)^2(3+x)^2(1-x)$
B) $y = -x^2(x-9)(x^2-3)$
C) $y = (x-6)(x-2)^2(x+2)^2$
D) $y = (x^2-1)^2(3-x)$

2 [MAT 2011 1A] A sketch of the graph $y = x^3 - x^2 - x + 1$ appears on which of the following axis?



(a) (b) (c) (d)

Chapter 4 · Original slide 15

Recap

15

If we sketched $y = (x-a)(x-b)^2(x-c)^3$ what happens on the x-axis at:

$x = a$:

$x = b$:

$x = c$: